9 **(a)** Charge distribution: Some of the alpha particles were observed to be scattered at large angles.
This shows that they encountered a positive charge exists in a very small volume known as the nucleus so that the coulomb repulsion is very large.
Mass distribution: Since the α particles are back scattered, the positive charge they collided with must also have a much larger mass, otherwise it would be the positive charges that were pushed away rather than the alpha particles scattered backward.
Hence, the mass of the atom also exists largely within a small volume (the nucleus)
Or

(a) Charge distribution: the positive charge of the atom exists in a very small space (the nucleus) at the centre of the atom.
Some (few) alpha particles are deflected by large angles.
Mass distribution: most of the mass of the atom exists in a very small space at the centre of the atom.
This is deduced from the fact that most alpha particles are able to travel through undeflected / Some alpha particles are deflected by large angles showing that the nucleus is more massive than alpha particles.

(b) **(i)** The probability of decay of a nucleus is unaffected by external factors (such as temperature, pressure or chemical composition).

(ii) The half-life of a radioactive nuclide is the time taken for the number of undecayed nuclei to be reduced to half its original number.

(iii) **1.** mass difference,

$$\Delta m = (239.052163 - 235.043930 - 4.003860)u = 0.004373\ u$$

$$\text{Energy released} = \Delta mc^2 = (0.004373)(1.66 \times 10^{-27})(3.00 \times 10^8)^2$$

$$= 6.533262 \times 10^{-13} \text{ J}$$

$$= 6.5 \times 10^{-13} \text{ J (shown)}$$

2.

$$E_{released} = E_U + E_\alpha = \frac{1}{2}m_U v_U^{\ 2} + \frac{1}{2}m_\alpha v_\alpha^{\ 2}$$

By principle of conservation of momentum,

$$0 = m_U v_U - m_\alpha v_\alpha$$

$$v_\alpha = \frac{m_U v_U}{m_\alpha}$$

$$E_{released} = \frac{1}{2}m_U v_U^{\ 2} + \frac{1}{2}m_\alpha \left(\frac{m_U v_U}{m_\alpha}\right)^2$$

$$v_U = \sqrt{\frac{2E_{released}}{m_U + \left(\frac{m_U^{\ 2}}{m_\alpha}\right)}}$$

$$= \sqrt{\frac{2(6.533262 \times 10^{-13})}{\left(235.043930 + \frac{235.043930^2}{4.003860}\right)(1.66 \times 10^{-27})}}$$

$$= 2.3684 \times 10^5 = 2.37 \times 10^5 \text{ m s}^{-1}$$

(iv) Alpha particles can easily be stopped by the outer layers of the skin due to their low penetrating power.

(c) **(i)** The number of undecayed nuclei decreases with time due to radioactive decay, and since activity is directly proportional to the number of undecayed nuclei ($A = \lambda N$), the activity decreases with time.

(ii) $A = A_0 e^{-\lambda t} \Rightarrow \ln A = -\lambda t + \ln A_0$

From the graph of ln A against t,

Gradient = $-\lambda$

$$= \frac{19.5 - 38.0}{(6.4 - 0.0) \times 10^5}$$

$= -2.8906 \times 10^{-5}$

$$t_{\frac{1}{2}} = \frac{\ln 2}{\lambda} = \frac{\ln 2}{2.8906 \times 10^{-5}}$$

$= 23979 = 2.40 \times 10^4$ years

(iii) Vertical-intercept = ln A_0 = 38.0

$A_0 = e^{38.0}$

$= 3.1856 \times 10^{16}$ year^{-1}

$A_0 = \lambda N_0$

$$N_0 = \frac{A_0}{\lambda} = \frac{3.1856 \times 10^{16}}{2.8906 \times 10^{-5}}$$

$= 1.1020 \times 10^{21}$

$= 1.10 \times 10^{21}$ (to 3 s.f.)

(d) $t_{\frac{1}{2},X} = 0.5 \times t_{\frac{1}{2},Pu}$

$\lambda_X = 2\lambda_{Pu}$

$A_{0,X} = 2A_{0,Pu}$

$\ln(A_{0,X}) = \ln(2A_{0,Pu}) = \ln(2) + \ln(A_{0,Pu}) = 0.69 + 38 = 38.69$

Straight line graph with twice the gradient as original graph

Vertical intercept is at 38.7 (approximately 38.5)